Akash Raj Patel

Computer Vision Application | Deep Learning | Smart Manufacturing

NTENSE Program Fellow and Smart Manufacturing Master’s candidate with a background in Computer Science. Expert in bridging AI and precision engineering for Digital Twin and industrial automation. Focused on architecting scalable, data-driven solutions for high-tech manufacturing.

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WORK EXPERIENCE

Corporate Sponsorship Fellow | HIWIN
International Industrial Talents Education Special Program

09/2027 - 09/2029 — *Taichung City, Taiwan.*

R&D Department, HIWIN Technologies, Taiwan.

Contact : 塗書怡上銀科技 - angel.tu@hiwin.tw

Graduate Research Student | CCU
Smart City Artificial Intelligence Research Lab

08/2025 - Present — *Chiayi County, Taiwan*

National Chung Cheng University

Achievements/Tasks

- Digital Twin Model Development
- RAG, QGIS, Unreal Engine 5 Simulation.

Contact : Pao-Ann Hsiung - pahsiung@ccu.edu.tw

Summer Research Intern | CCU
India Taiwan Joint Research Center For AI

07/2023 - 10/2023 — *Chiayi County, Taiwan*

National Chung Cheng University

Achievements/Tasks

- Led a team of 6 interns from India, Taiwan and Denmark in a computer vision project in collaboration with the Taiwan Fisheries Association.
- Developed an End-to-end system that was able to detect and log how many overtime hours fishermen on board a fishing vessel worked just from a given CCTV Footage.
- Successfully incorporated custom-tuned YOLOV8 model and data preprocessing. Used Tensorflow, ScikitLearn, PyTorch, Keras, Yolo, among other libraries.
- Submitted a comprehensive 60 page research report on the topic.

Contact : Pao-Ann Hsiung - pahsiung@ccu.edu.tw

EDUCATION

Masters In Advanced Manufacturing
AIM-HI, National Chung Cheng University,

08/2025 - Present — *85%*

Bachelors In Technology, Computer Science
Amity University, Mumbai, India.

08/2021 - 08/2025 — *8.74GPA*

SKILLS

Data Visualization | Machine Learning | Deep Learning | Statistical Analysis | Computer Vision | Collaborative Problem Solving | Leadership Skills | Technical & Business Presentation

AWARDS & ACHIEVEMENTS

International Industrial Talents Education Special Program (INTENSE Program) Fellow At Hiwin Technologies (09/2025 - Present)

- Recipient Of the Prestigious Taiwan Government-Industry-Academia Fellowship for the recruitment of international superior industrial talents students and cultivating high-quality professionals with international mobility.
- Receives full tuition scholarship by the National Development Fund.
- 10,000 NTD(300USD) Monthly living allowance paid for by the partner company (HIWIN Technologies).

Receiver Of Dr. Ashok K. Chauhan 100% Scholarship (08/2020 - 08/2023)

- Was Awarded Dr.Ashok K. Chauhan 100% Merit Scholarship for the tuition of my 4 year Bachelors Program at Amity University.

2nd Rank, NIRMAN HACKATHON 2023 - Amity University Mumbai (02/2023 - 02/2023)

- Achieved 2nd Rank In NIRMAN 2023 and a cash prize of 30,000 INR (450 USD), 48 Hours Hackathon hosted at amity university Mumbai.
- Designed A python flask API to convert a passed 2D ArcGIS.shp file into a 3D model And render it on a webApp.

PROJECTS

Image Color Restoration Using AutoEncoders - Python

A Python TensorFlow project for image colorization using a pix2pix GAN deep learning model, allowing users to upload and colorize grayscale images Enhanced

Neural Artistic Style Transfer | CPU-Only Optimized for Real-Time Web Application - Undergraduate Thesis. (05/2024 - 06/2025)

Developed a Fast Neural Artistic Style Transfer (NAST) model, trained exclusively on a CPU-only setup, highly optimized to be deployed as a real-time web application on servers without dedicated GPUs.

2D Satellite Data into 3D City Models API Flask WebApp – Python | (NIRMANN 2023 Winner)

A python flask API to convert a passed 2D ArcGIS.shp file into a 3D model And render it on a webApp. Utilized native PyOpenGL libraries to achieve fast and efficient performance and conserve server memory